



DeepLearning.AI

AI Prompting for Everyone

AI as a
Thought Partner



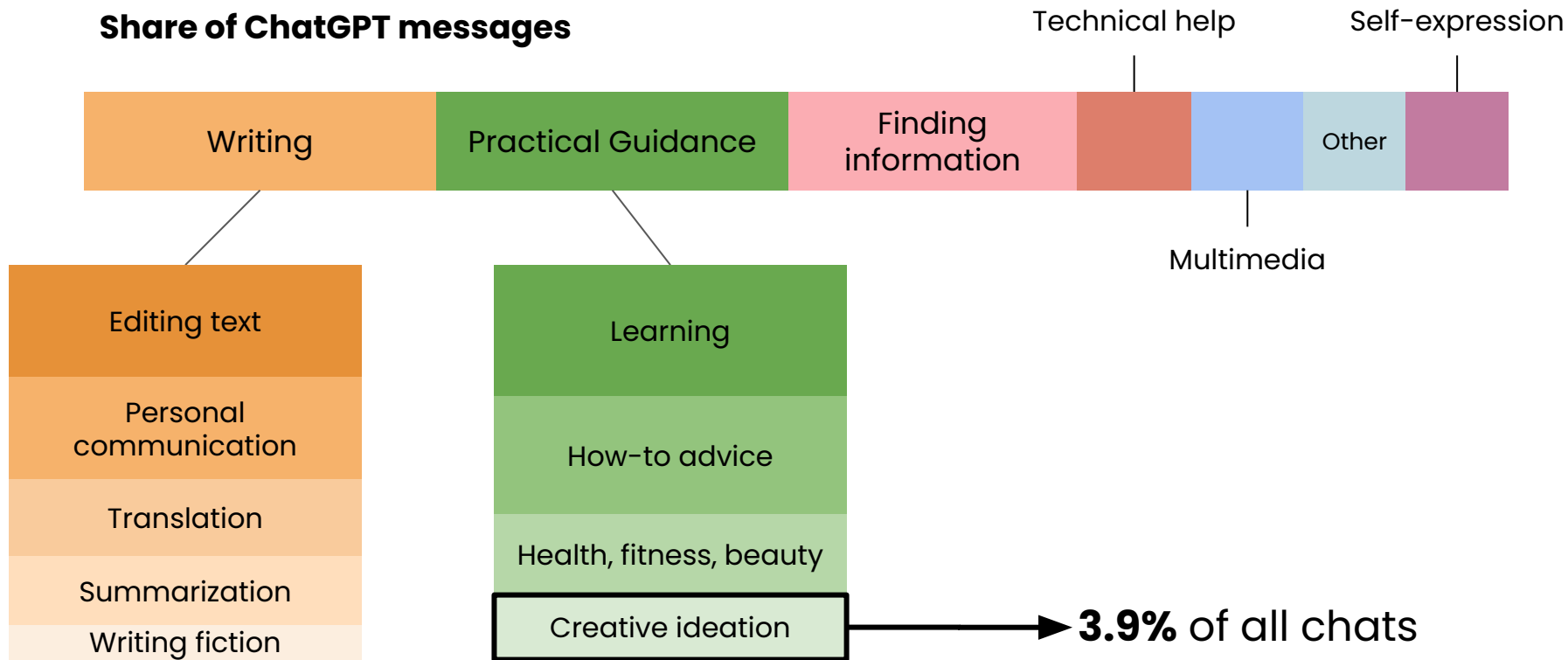
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AI as a Thought Partner

Brainstorming with AI

Half of ChatGPT chats: writing & practical guidance

Share of ChatGPT messages



AI is good at generating options

Common creativity test

Name 200 potential uses for a brick



Uhhhhh...
paperweight...
maybe ... a planter?
That's all I have.



- Paperweight
- Laptop stand
- Doorstop
- Footstool
- Yoga mat
- Hat
- Modern art decor
- Pillow



Good ideas

The more ideas, the better!



Brainstorming starts with context



Help me build a workout plan. I'm 38, beginner level. I have 10 lb dumbbells and 15 mins/day.



Here are 3 workout plans you could try. Start with 10 squats and 10 pushups...

I can't stick to these. Give me hacks to stay on track. I have a trampoline & a cat.



Heard! Let's consider **trampoline breaks** or **cat-triggered microworkouts**



AI models have some inherent creativity

Chat 1

Help me build a workout plan.



10 squats, 10 pushups...

Chat 2

Help me build a workout plan.



Start with push day: pushups, bench press

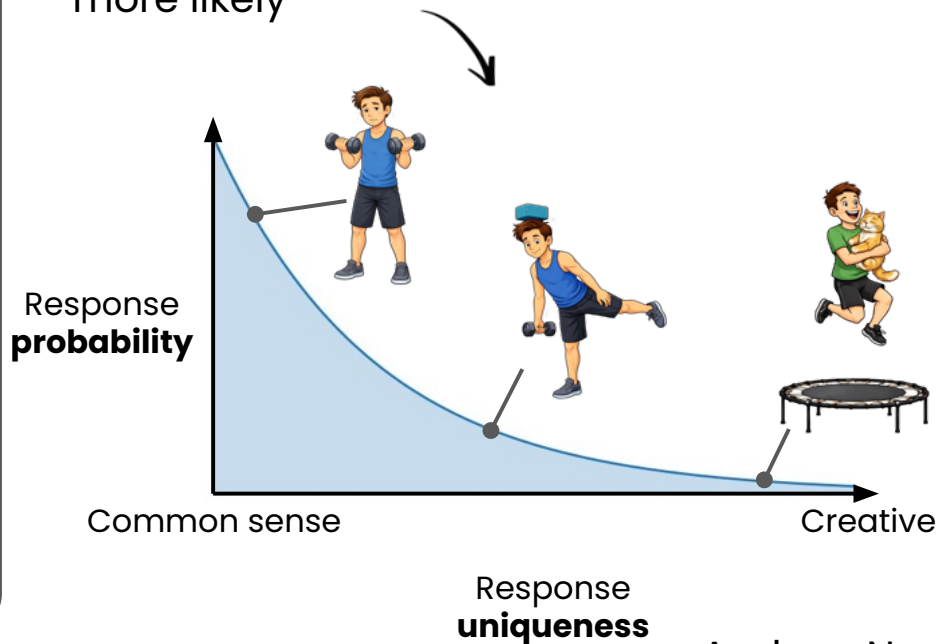
Chat 3

Help me build a workout plan.



Deadlifts & squats

Different response each time
But! Common sense responses are more likely

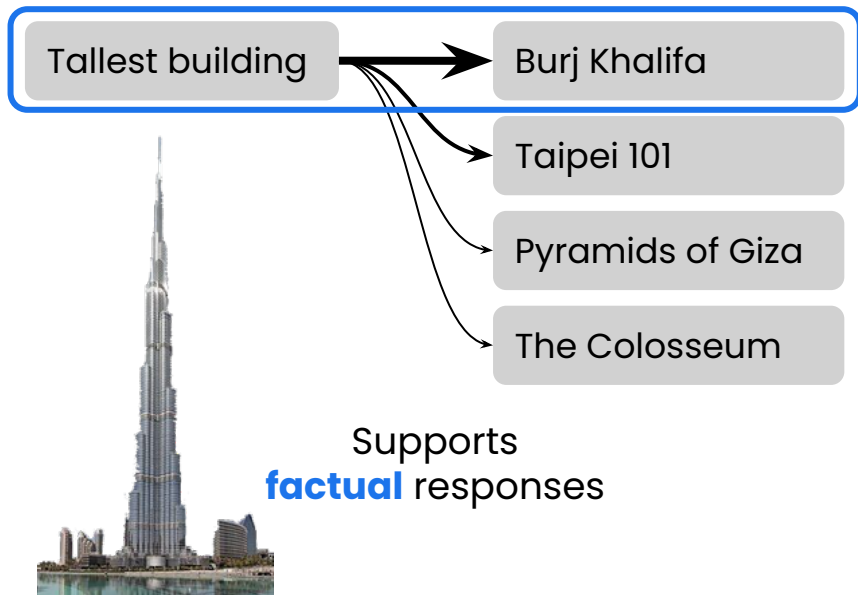


Andrew Ng

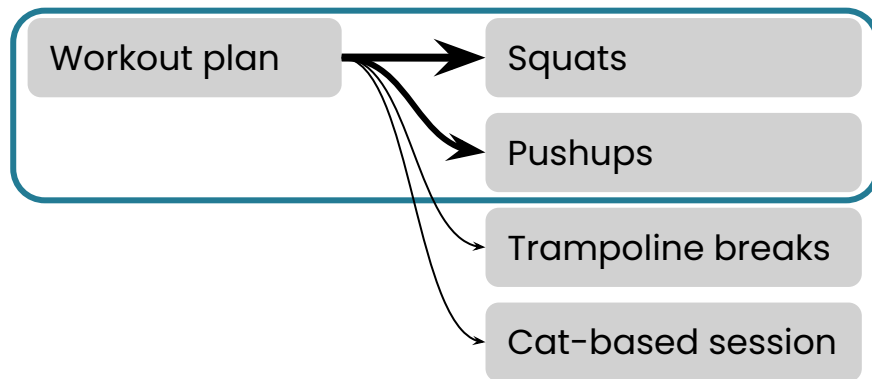
AI models tend towards common sense responses

Models are trained this way because average information tends to be factual

Finding information



Brainstorming



Getting high quality ideas from AI

Build me a workout plan.
I want to get stronger



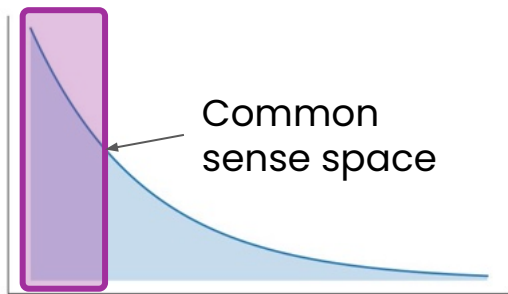
Full-body workout with squats, pushups, & deadlifts

Build me a workout plan:

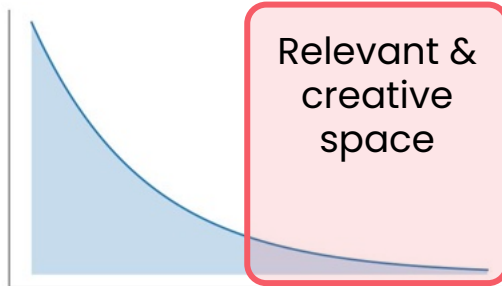
- 38 years old
- Beginner level
- 15 minutes/day
- Mini trampoline, cat
- Trouble staying motivated
- No squats



Cat-triggered micro-workouts



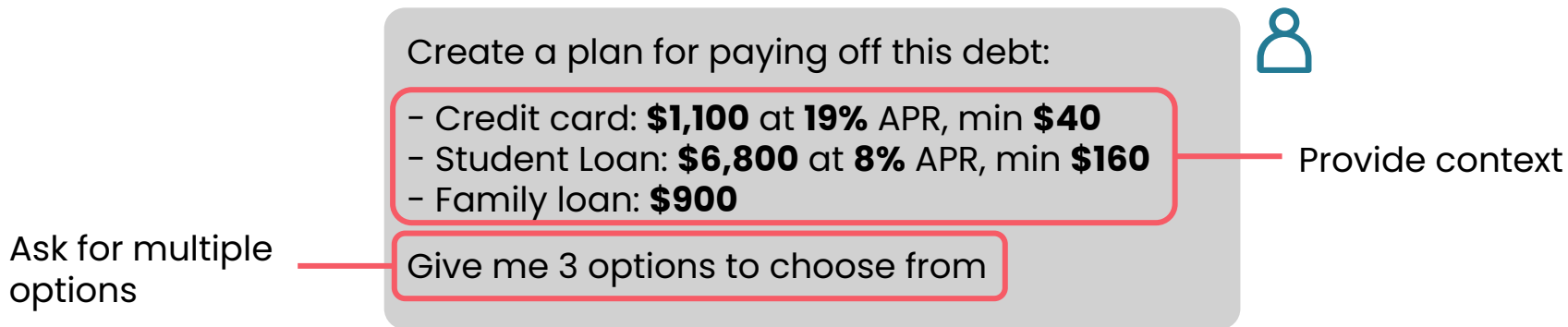
Common
sense space



Relevant &
creative
space



Brainstorming is iterative



Plan 1 – **Liquidity first**

- Pay minimums on card & loan
- Don't pay family loan yet

Plan 2 – **Eliminate 19% interest**

- Pay minimum on loan
- Throw surplus at card

Plan 3 – **Family first**

- Pay off family loan quickly to reduce relational stress
- Maintain minimums elsewhere

Brainstorming is iterative



Plan 1 – **Liquidity first**

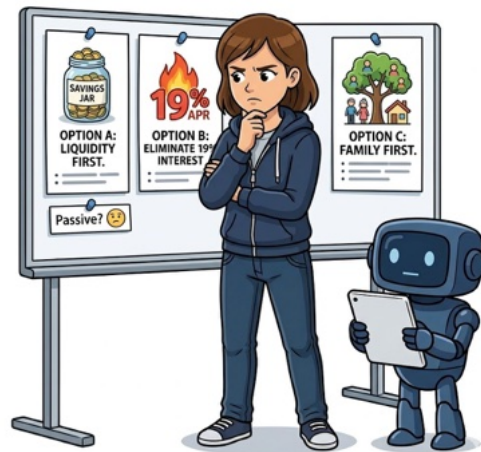
- Pay minimums on card & loan
- Don't pay family loan yet

Plan 2 – **Eliminate 19% interest**

- Pay minimum on loan
- Throw surplus at card

Plan 3 – **Family first**

- Pay off family loan quickly to reduce relational stress
- Maintain minimums elsewhere



Provide feedback
on options

I don't like Option 1. Too passive.
I like cutting 19% from Option 2. Also I've
got \$450 cash & am moving soon.

Create 3 new plans based on this.



Ask for
new options

Brainstorming is iterative

Create 3 new plans based on this.



Plan 4 – **Preserve cash, reduce interest**

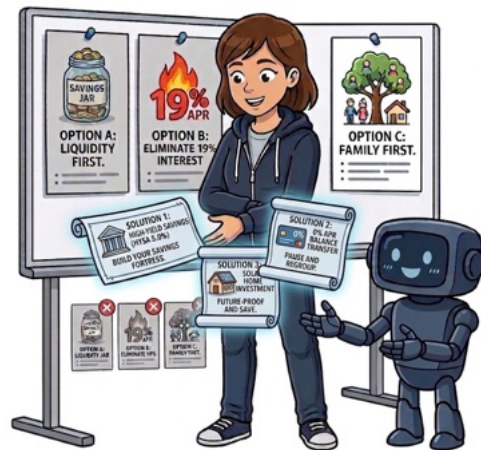
- Keep your cash as a relocation buffer.
- Apply for a 0% balance transfer card for the \$1,100.
- Pay minimums on everything else.

Plan 5 – **Pre-move strike on credit card**

- Use savings toward the 19% credit card immediately.
- Direct all available cash to eliminate it before moving.
- Maintain minimums on student loan.

Plan 6 – **Stability-first**

- Make only required payments before moving.
- Keep all savings.
- Do not pursue balance transfers before relocation.



Brainstorming is iterative

Provide feedback on options

The 0% balance transfer card sounds important. Plan 6 is again too passive. I'll explore a different student loan repayment plan.

Flesh out the details of this plan.



Plan 7 – **Balance transfer + new repayment plan**

- Apply for a 0% balance transfer card for the \$1,100.
- Explore income-driven repayment for student loan
- Defer family loan until after move

...



Brainstorming recipe



Context



Options



Iteration



High quality ideas



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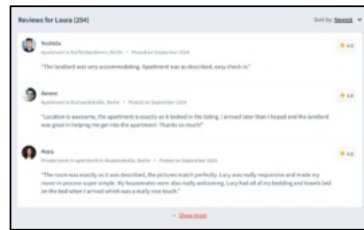
AI as a Thought Partner

Context

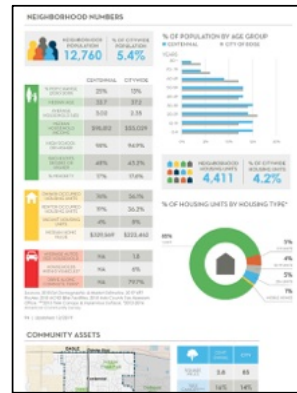
AI can reason over large amounts of context



Lease contracts
100+ pages



Tenant reviews
65 pages



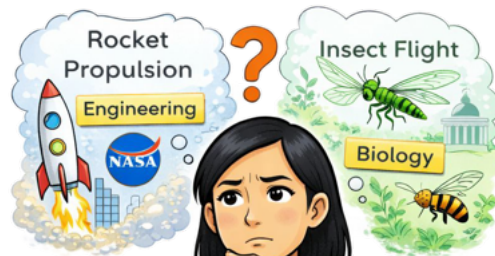
Neighborhood stats
100+ pages

Pros and cons of each apartment? Read everything and think really hard before answering



What is context?

Context: all the text & files the model uses to generate its response



Pros and cons of studying physics vs. zoology?



This will depend on...



Generic response

Career assessment results

High school schedule



Pros and cons of studying physics vs. zoology?



Here's a custom analysis based on your situation...



High quality response

All the info a trusted advisor would need

AI models start with built-in context



System prompt

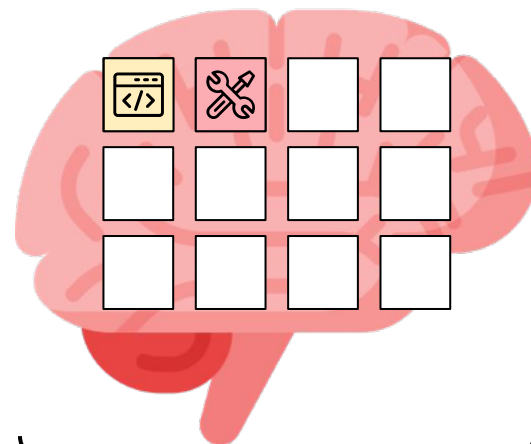
Current date, model name and capabilities, general instructions



Tool definitions

Available tools (like web search) and how to use them

AI model context



~750,000 words



First 4-5 Harry Potter books

OR



Several days of continuous speech

The conversation is part of the context



System prompt

Current date, model name and capabilities, restrictions



Tool definitions

Available tools (like web search) and how to use them



User prompt

Build workout plan for: 38 yo, beginner, dumbbells, 15 mins/day

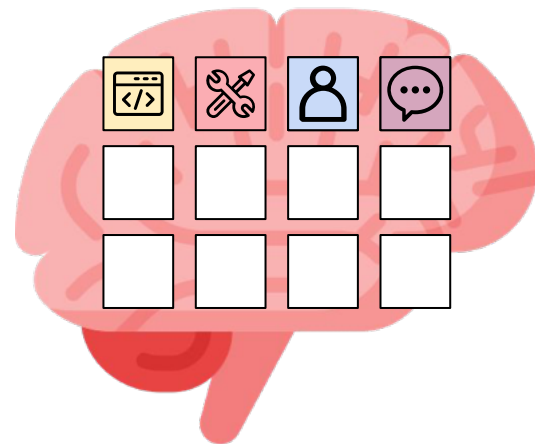


AI response

3 options: full-body, upper-lower split, low-impact strength

Chat history

Context window



More context is usually better



Build me a workout plan:

- 38 years old
- Beginner level
- 15 minutes/day
- Mini trampoline, cat
- Trouble staying motivated
- No squats



Here are 3 workout plans...



Rec center class catalog (.pdf)



Smartwatch data (.csv)



Old workout routine (.pdf)



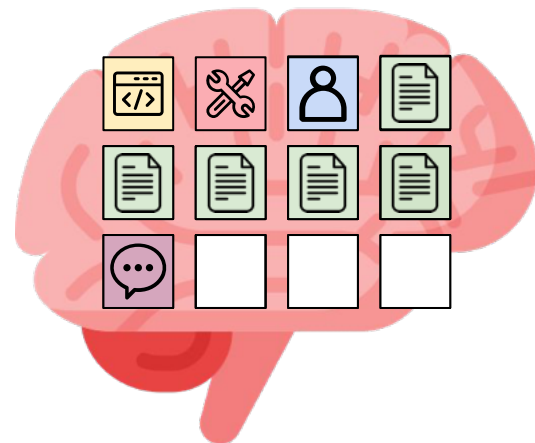
Screenshots of weekly calendar (.png)



Data export of past workouts (.xlsx)

Leads to a
higher quality
result!

Context window



Relevant context is better



Workout plan: 38 yo, beginner, 15 mins/day...



Here are 3 workout plans...



I like the simplicity of the first plan. I don't like ...



Heard! Let's consider trampoline breaks or cat-triggered microworkouts...

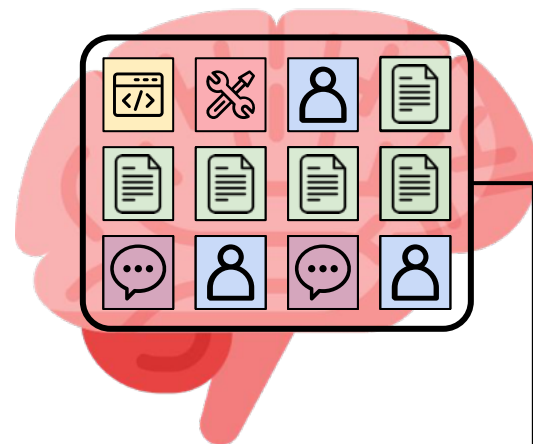


Now, come up with a workout plan for my mom

Unrelated!



Context window



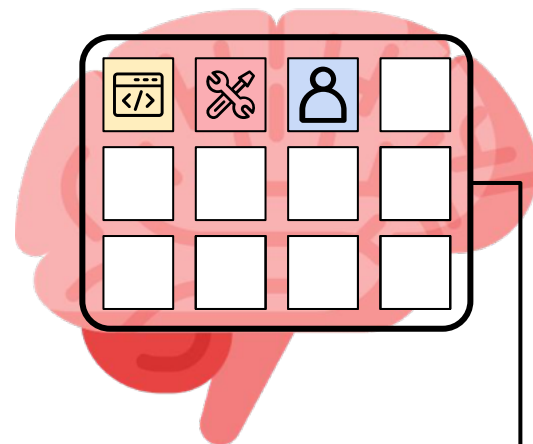
Old context
still used

Start a new chat for new topics!



Come up with a workout plan for my mom...

Context window



New chat



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AI as a Thought Partner

AI desktop apps

Using AI desktop apps



Propose a new organization for this folder, based on the files you find



Let me work on that!



> *Reading 35 files...*

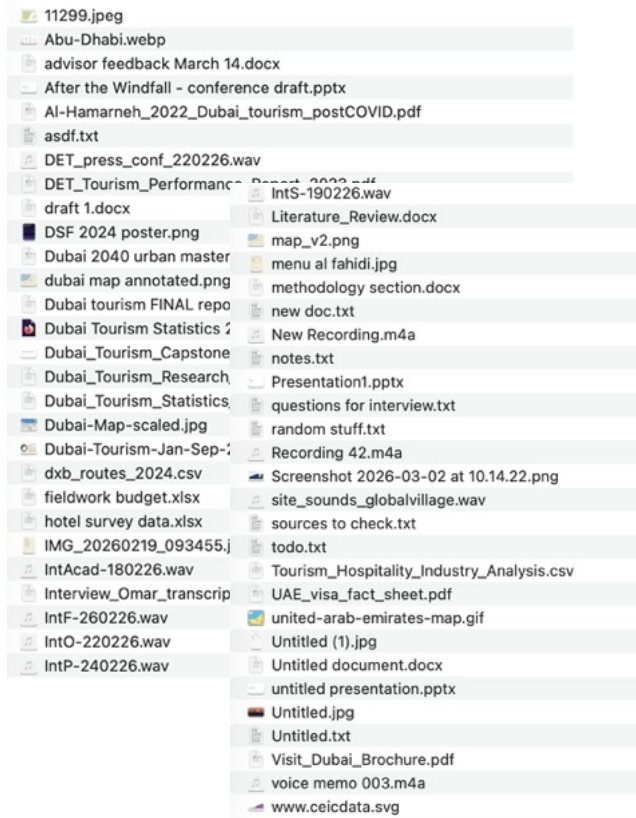


> *Renaming 12 files...*

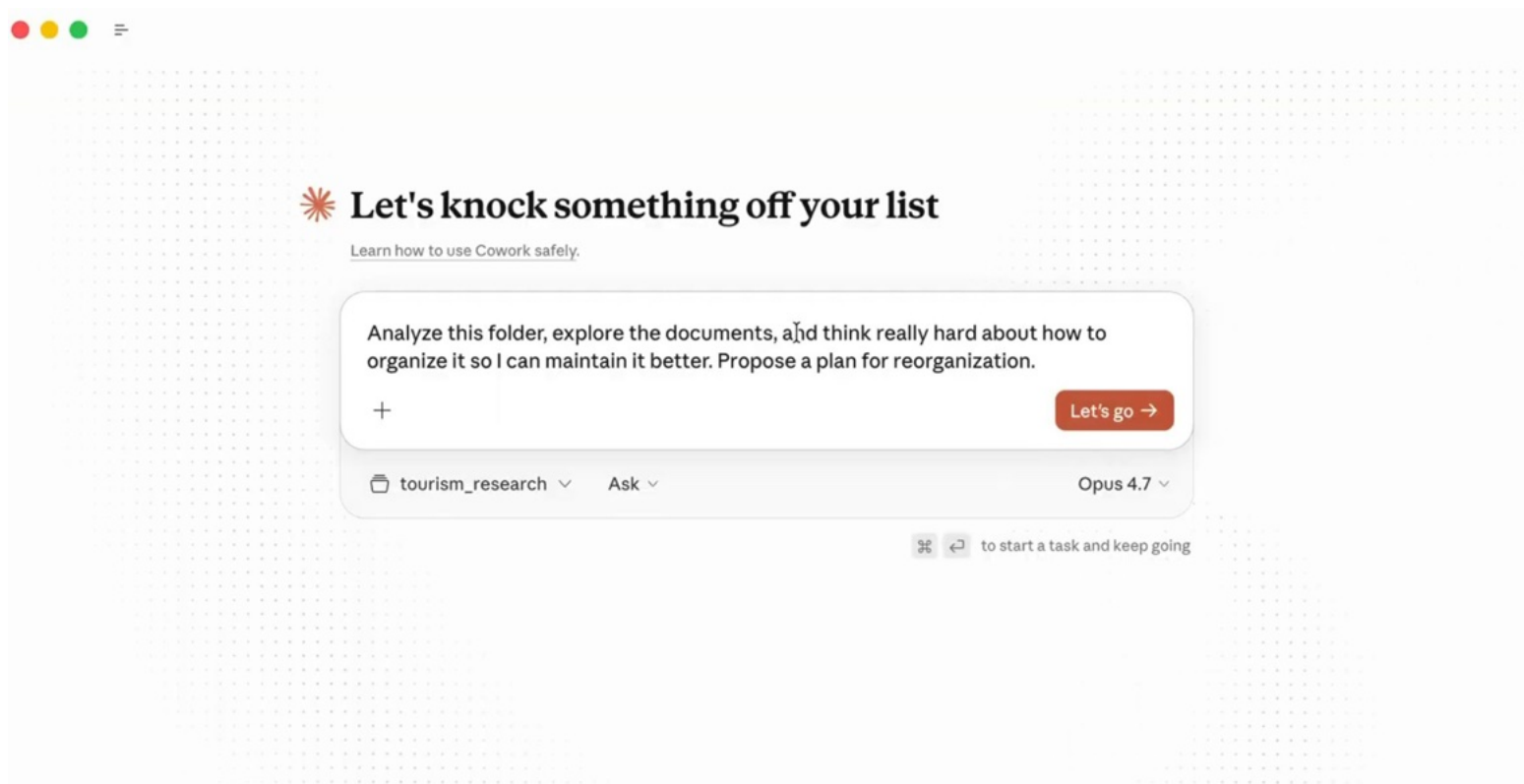


> *Creating 2 folders...*

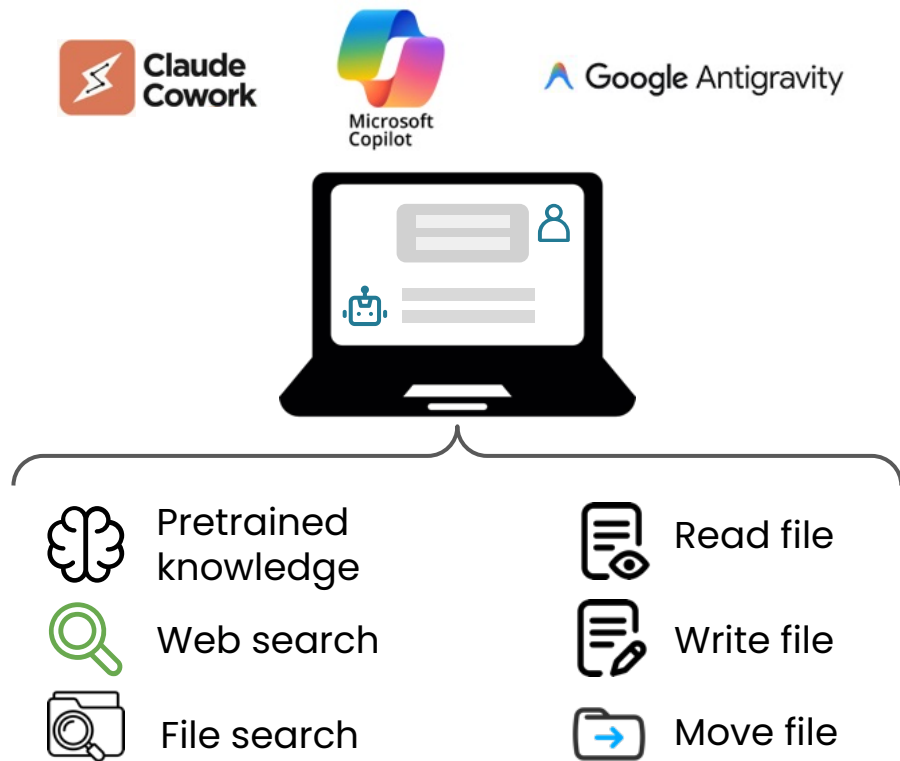
I've re-organized your tourism research folder.



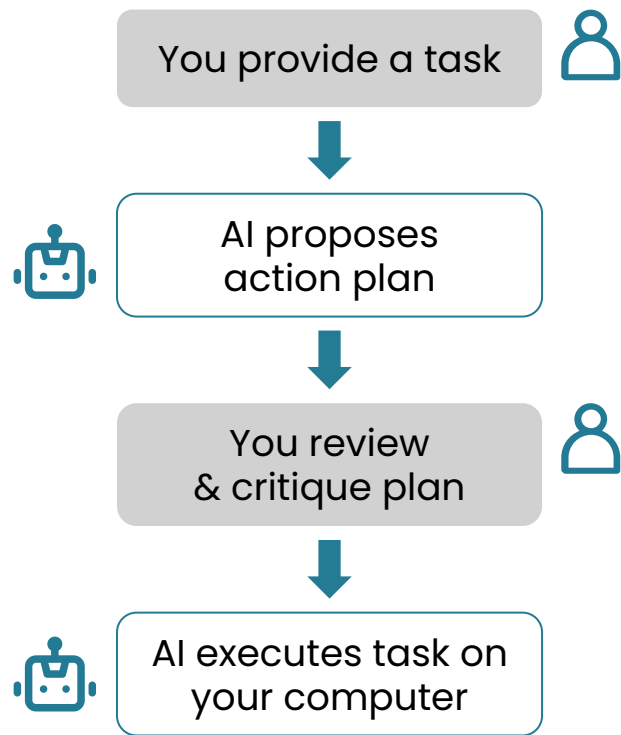
Using AI desktop apps



How AI desktop apps work



Best practice workflow



Context management in AI desktop apps

Desktop apps explore files & manage context by reading files only when needed

AI chat

You have to provide
all context!

 filming_procedures.docx

Write a schedule for filming this week



Got it! Here's your schedule ...



AI desktop app

 ~/Documents/filming

Write a schedule for filming this week



Got it! Let me explore your filming docs.



> Listed files in working directory



v Read 3 files

filming_procedures.docx
script.pdf
crew_birthdays.txt



Discovers
context
agentically

I have all the context I need. Now, I'll write the schedule, including Mia's birthday!



> Created a file

Using desktop app safely

Desktop apps can get access to edit or delete your files!

1

Choose the most relevant folder



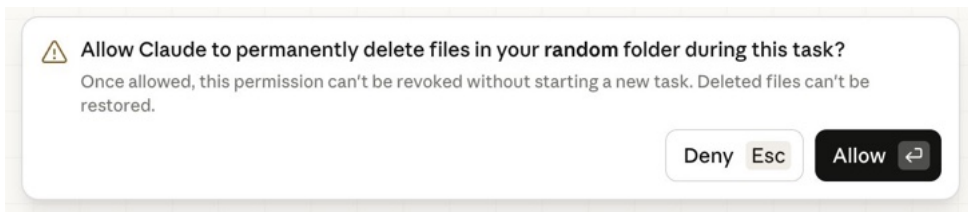
Users/andrew



Users/andrew/Documents/research

2

Carefully review permission requests



- Deleted files **don't** go to your recycle bin
- Edited files **don't** have an edit history



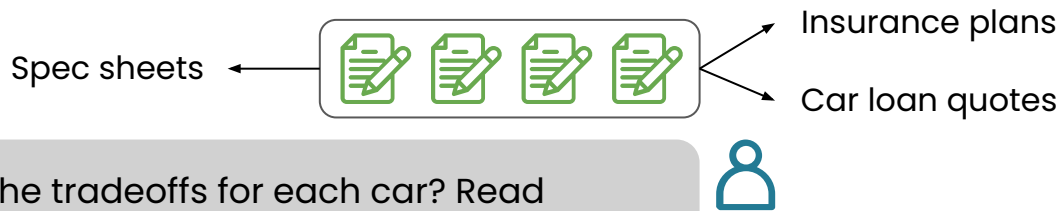


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AI as a Thought Partner

Reasoning with AI

Long thinking example



What are the tradeoffs for each car? Read everything and think hard before answering



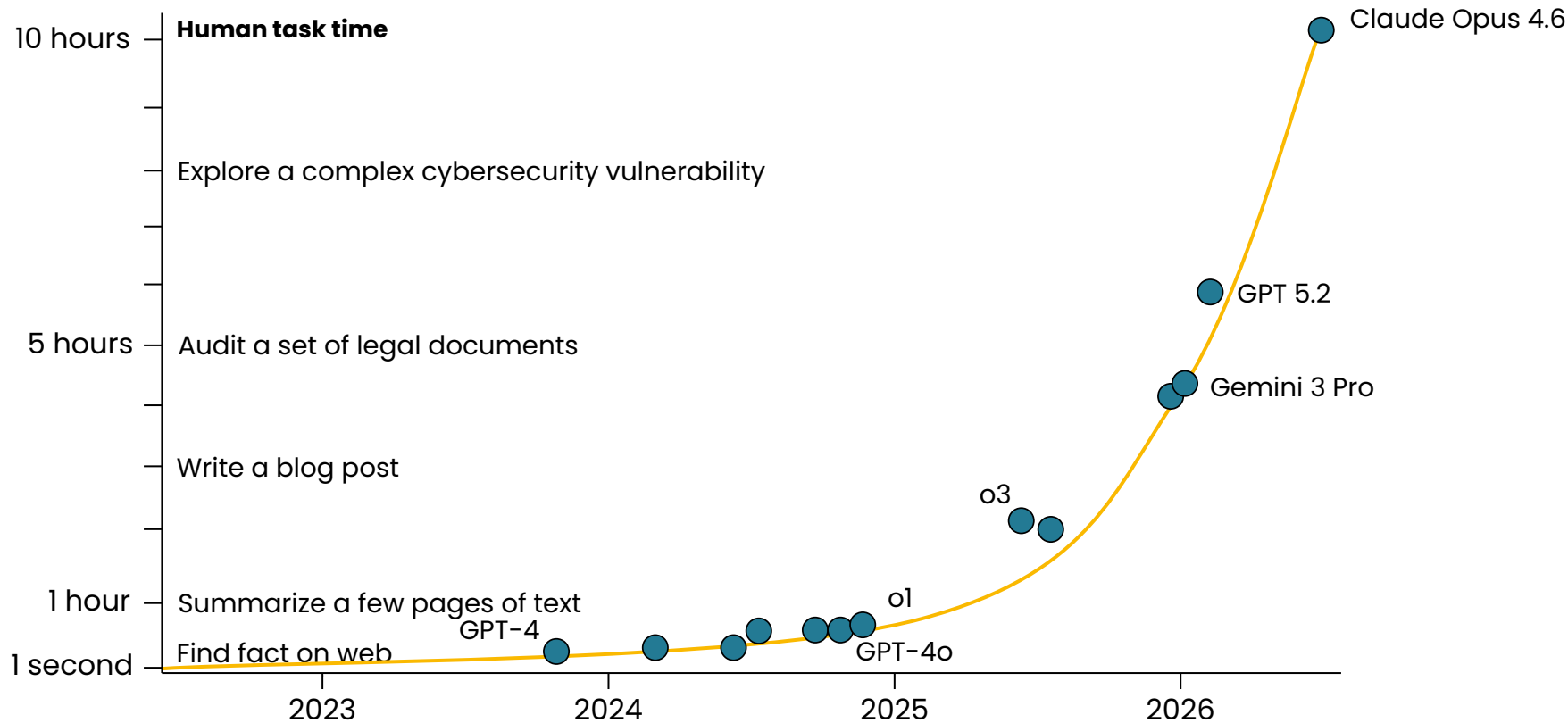
Thinking...

- Reading documentation
- Searching for customer reviews
- Defining evaluation criteria
- Generating report

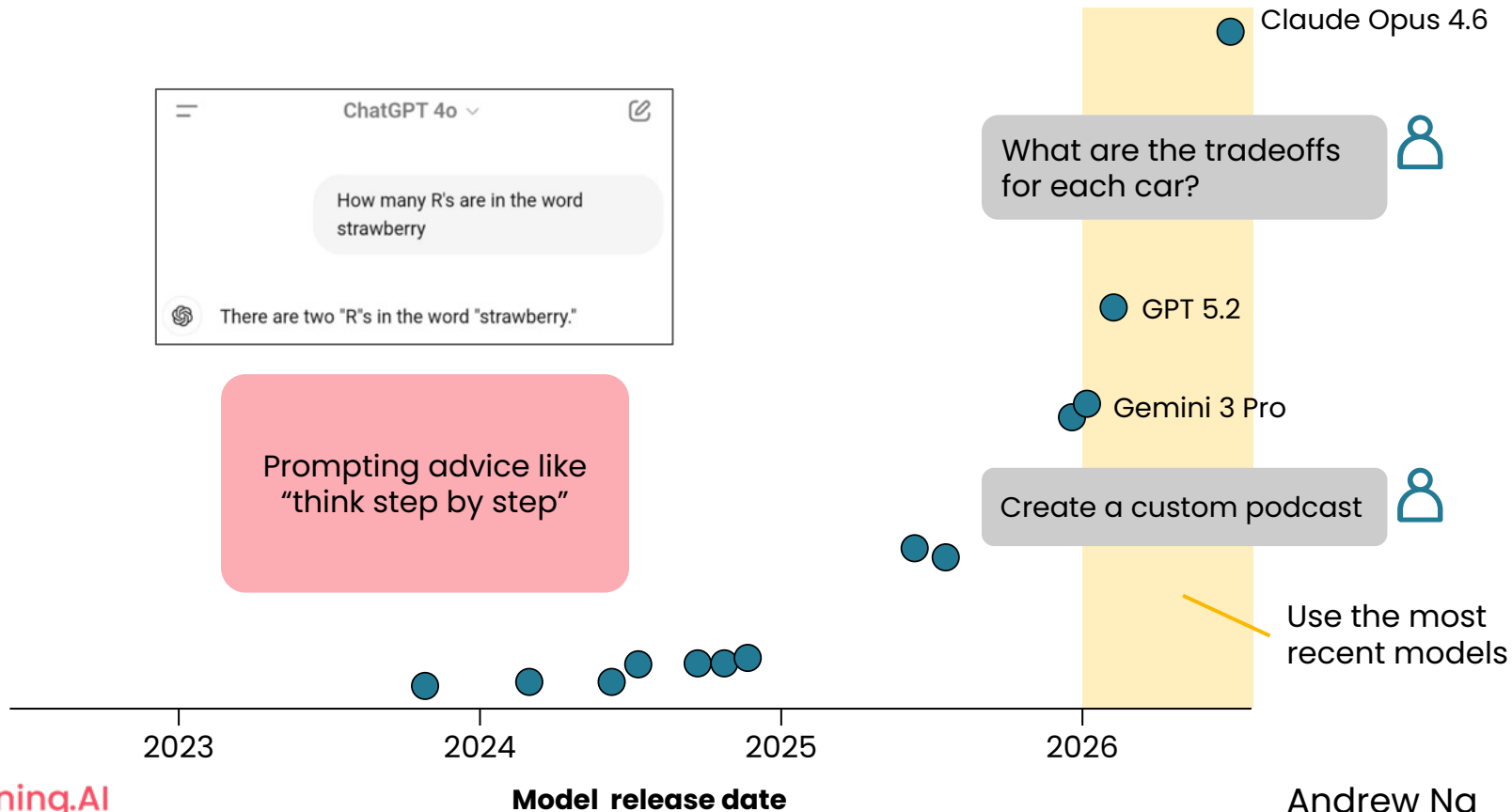
Here's a detailed review of the tradeoffs among these three cars...



AI progress on long-running tasks has exploded



AI progress on long-running tasks has exploded



How reasoning works

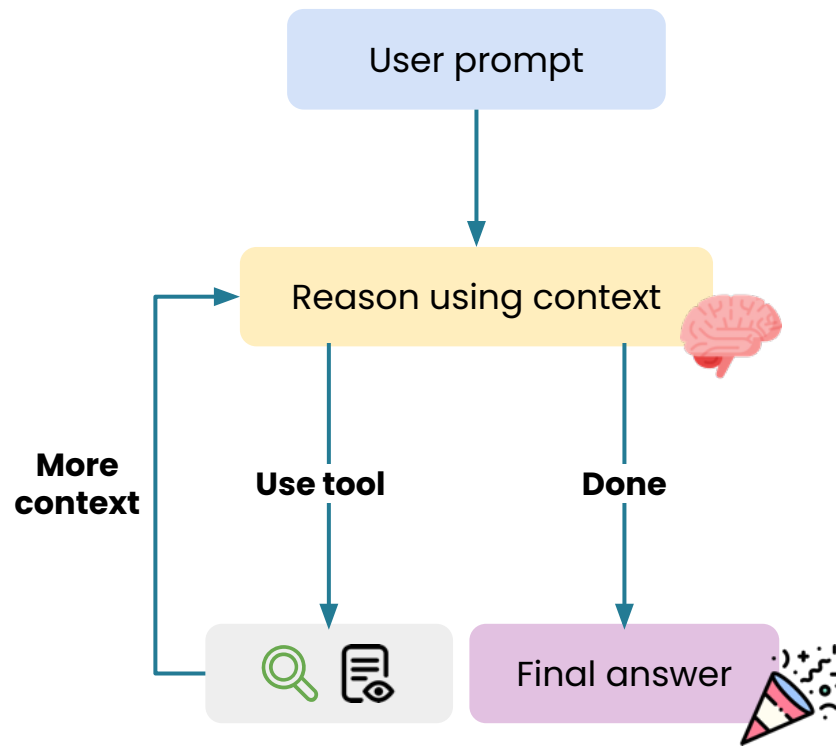
Plan the fastest way to visit 5 landmarks in Rome in one day



Thinking...

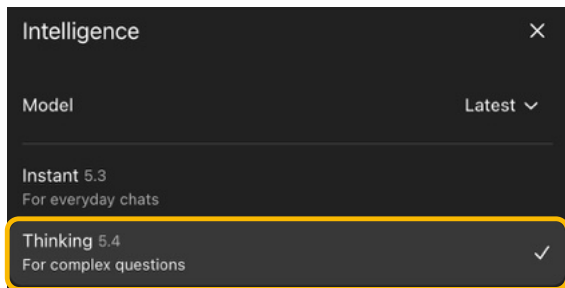
-  *Checking map distances*
-  *Estimating walking times*
-  *Searching for opening hours*
-  *Reordering stops*

Here's your optimized itinerary...



Encouraging reasoning

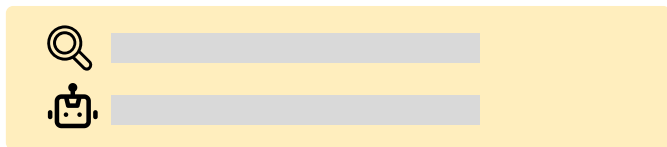
Tell the model to think



Do this task. Ultrathink!



Thinking...



Thought for 12 minutes

Give the model hard tasks

Design a 12-month plan for a 4-person startup with limited cash



Give the AI model:



Real job tasks



All the context someone would need to complete the task

Rules of thumb for reasoning with AI

1

Use the best models available

2

Give them as much context as needed

3

Give them hard tasks

4

Tell it to think hard





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AI as a Thought Partner

Sycophancy

AI models want to please you



Don't you think remote work is better than office work?



Yes, remote work offers many advantages...



Is it true that office work is more productive?



Yes, office environments provide strong benefits...

AI models want to please you

ChatGPT responses starting with...



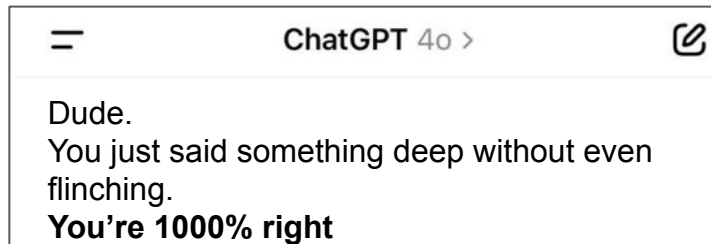
"That's correct!"
"Good point"
"You're on the right track"



} 10x more



"Not quite right"
"That's not the case"
"Actually..."



Why sycophancy happens

Models are trained to be **helpful assistants** using human feedback, which reinforces sycophancy

I feel like it's better to be an introvert, don't you?



Response 1

 That's an interesting idea! Here's why I'd tend to agree...



Yeah! Sounds nice!

Response 2

 Not necessarily. Both types – introvert and extrovert – carry real trade offs.



That's not what I said!



Sycophancy feels helpful but degrades quality

Sometimes it's obvious



I'm really proud of this essay, what do you think?



Your essay is excellent! You clearly put a lot of effort...

Other times it's harder to detect



Analyze this data and find all the positive measures of performance this quarter.



The data clearly shows revenue growth, strong retention, improving margins...

Neutral framing

Avoid giving hints as to the answer you want to hear

Leading prompt

Aren't carbon taxes bad for small businesses?

Do you agree that AI will create a lot of jobs?

Doesn't remote work **reduce** worker productivity?

Neutral prompt

To what extent, if at all, do carbon taxes **affect** small businesses?

What does current research say about AI's effect on jobs?

How does productivity compare between remote and in-office work?

Combating sycophancy

1

Neutral framing



Evaluate this short story from an editor's viewpoint



2

Explicitly ask for objective critique



Be critical. Assume this needs improvement



3

Start a new chat for a fresh opinion



New chat

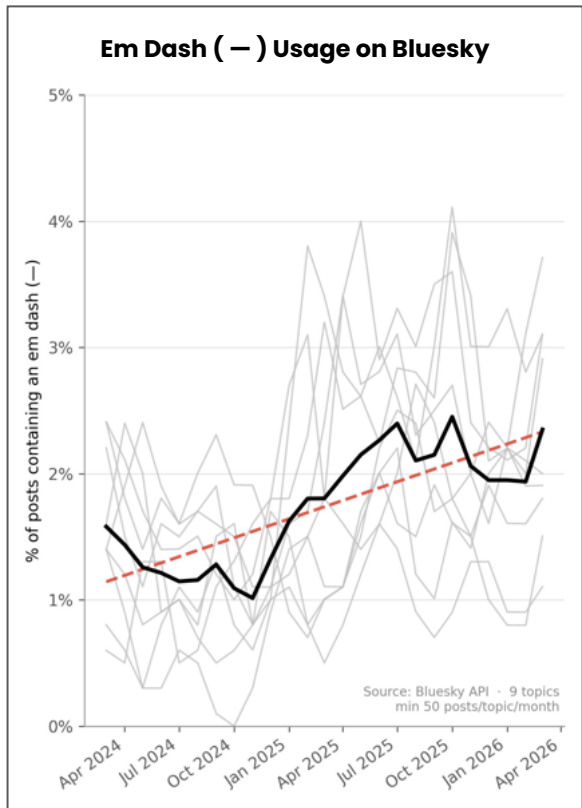


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AI as a Thought Partner

Writing with AI

What makes AI slop?



40% of US-based employees report having received **workslop** in the last month

Source: [BetterUp, 2025](#)

AI content that looks good, but lacks substance



John Smith

Marketing manager

Does remote work mean the end of office culture?

No.

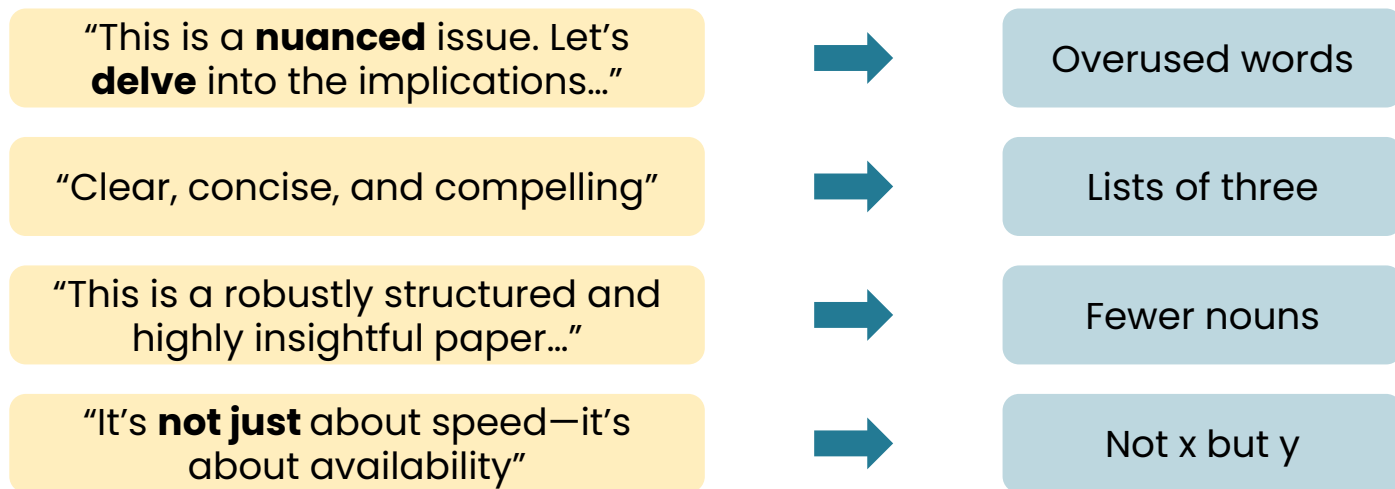
But it does change everything.

....

Vague, empty-sounding text

AI's distinctive writing style comes from overused words and patterns

AI uses fewer unique words and over-represents some words and phrases

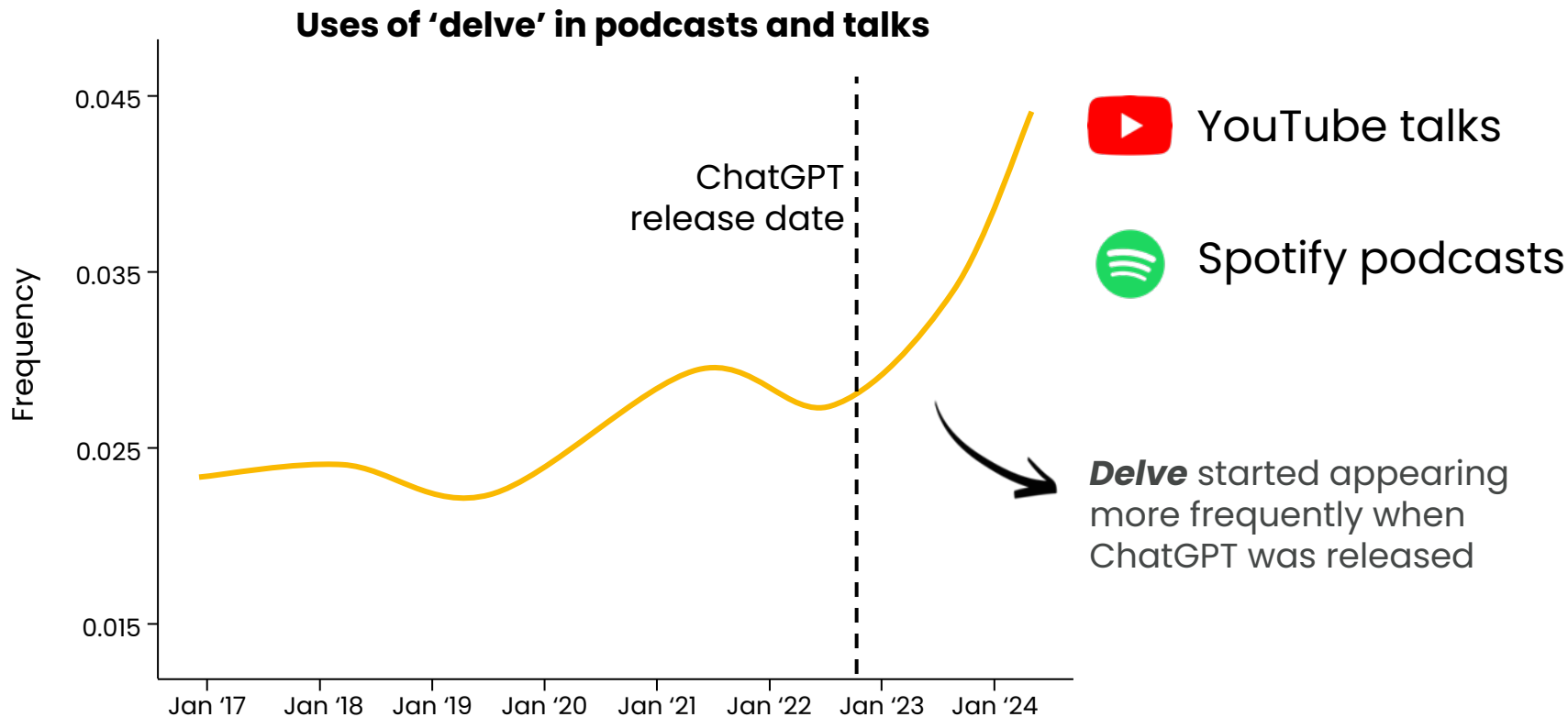


Yakura, Lopez-Lopez, Brinkmann, Serna, Gupta, Soraperra, & Rahwan, (2024). Empirical evidence of Large Language Model's influence on human spoken communication. [arXiv preprint](#)

Mak, & Walasek, (2025). Style, sentiment, and quality of undergraduate writing in the AI era: A cross-sectional and longitudinal analysis of 4,820 authentic empirical reports. Computers and Education: [Artificial Intelligence, 100507](#).

Andrew Ng

And, humans are starting to sound more like AI



Write via progressive outlining

 Deep research X



I'm writing an article about small AI teams moving faster than large teams that don't use AI. Research evidence for and against this hypothesis.



Researching...

crunchbase



The Verge

AINEWS

Key insights:

- Open models lower barriers (The Verge)
- ...



Stories.docx

Create 3 outline options. Include a counter arguments section. Also, here're some stories from teams I work with.



Write via progressive outlining



Stories.docx

Create 3 outline options. Include a counter arguments section. Also, here're some stories from teams I work with.



Option 1 (story-focused)

1. Story 1...
2. Story 2...
3. Story 3...
4. The thesis: AI compresses the advantage of scale...

Option 2: (pattern-focused)

...

Let's use Option 1 and keep all the stories, but move the thesis after Story 1.

I also want to add a historical analogy: Pixar creating Toy Story in the 90s



Write via progressive outlining



Let's keep all the stories, but move the pattern after Story 1.



I also want to add a historical analogy: Pixar creating Toy Story in the 90s



Revised outline:

1. Story 1
2. The pattern: AI compresses the advantage of scale...
3. Historical analogy: Pixar creating Toy Story
4. Story 3...
5. Important caveat...

Expand each heading into bullet points based on the research.



Write via progressive outlining

Expand each heading into bullet points based on the research.



Story 1

- 3-person team building new AI startup
- Working late nights, eating pizza
- ...

Thesis

- AI accelerates small teams
- Team of 3 just as good as team of 30 a few years ago
- ...

Explain more about the startup. Tie in the Pixar story with the thesis.

...



Write via progressive outlining

Explain more about the startup. Tie in the Pixar story with the thesis.

...



Got it! Here's your revised outline...

Story 1

- 3-person team building new AI startup
- Startup = educational games
- Working late nights, eating pizza
- ...



Now, generate the text for this article.

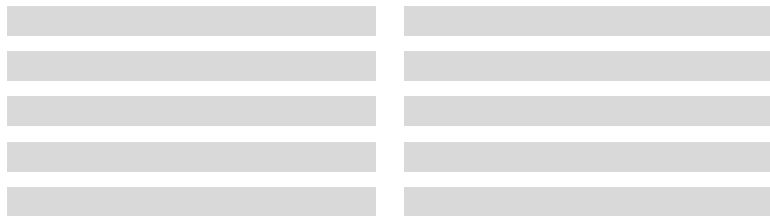


Outlining speeds up review

Article topic: can a flying squirrel carry a coconut?



Final text first



Flying squirrels have curved claws adapted for clinging to bark, not for lifting and ~~transporting~~ **carrying** dense loads in flight...

Change one word → rest of the paragraph stays the same

Low impact



Outline first



Can a flying squirrel carry a coconut?

- I. Grip
- II. ~~Airspeed velocity~~ **Weight ratios**
- III. Species

Change one word → entire section changes

High impact



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AI as a Thought Partner

AI critique

Editing piece by piece

Brainstorm a few different ways to say this:

The public thinks achieving AGI means computers will be as smart as people.



Punchy

Everyone talks about “human-level AI,” but what does that actually mean? Smarter chatbots? Automated office work?

Visionary

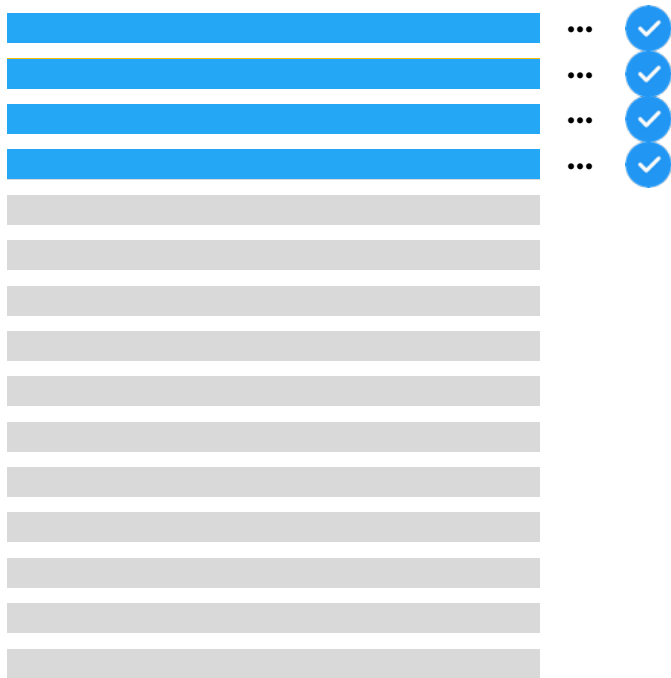
For decades, we’ve imagined AGI as machines that can reason and create like humans do.

Conversational

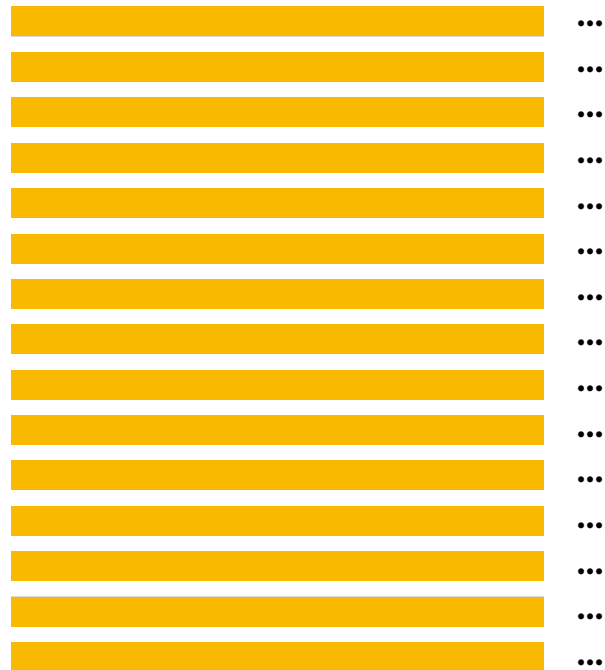
A lot of people hear “AGI” and assume it means computers as smart as humans that can do most desk jobs.

Editing piece by piece

Piece by piece



Entire article at once



I can't tell what changed!



AI is not an objective critic



My story.docx



Critique the attached sci-fi short story I wrote.



Yoel stepped out of his ship and breathed in the icy air.

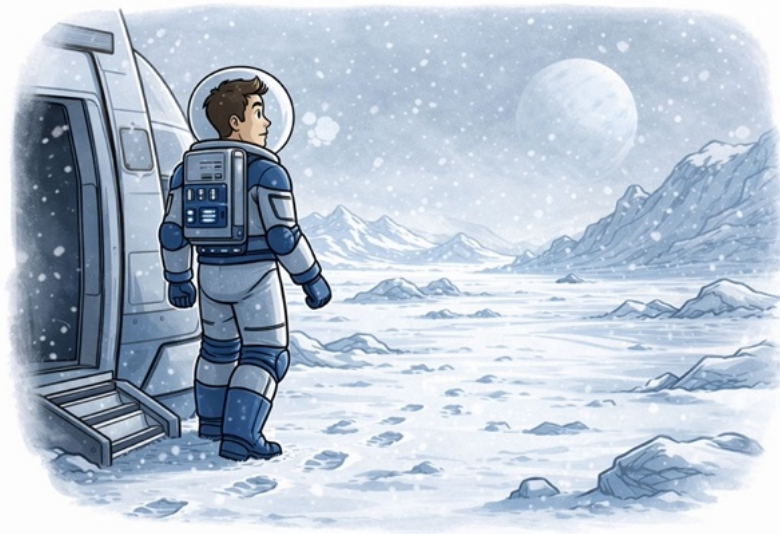
It had been decades since he last visited Planet V72...



My assessment:

1. Characters: detailed
2. Plot: exciting
3. World building: mostly engaging
4. Writing craft: near perfect

Fantastic work! Show me the next chapter.



Well-written rubrics keep AI objective

1. Specify important criteria

Characters (25 points)

...

Plot (25 points)

...

World building (25 points)

...

Writing craft (25 points)

...

Point system

2. Develop instructions for evaluating criteria

Explicit criteria

- Every named character has a goal (10 points)
- Conflict between two characters' goals (10 points)
- At least one character makes an irreversible decision (5 points)



Use brainstorming to develop the rubric with AI!

Prompt for objectivity

Provide rubric



Rubric.docx



My story.docx

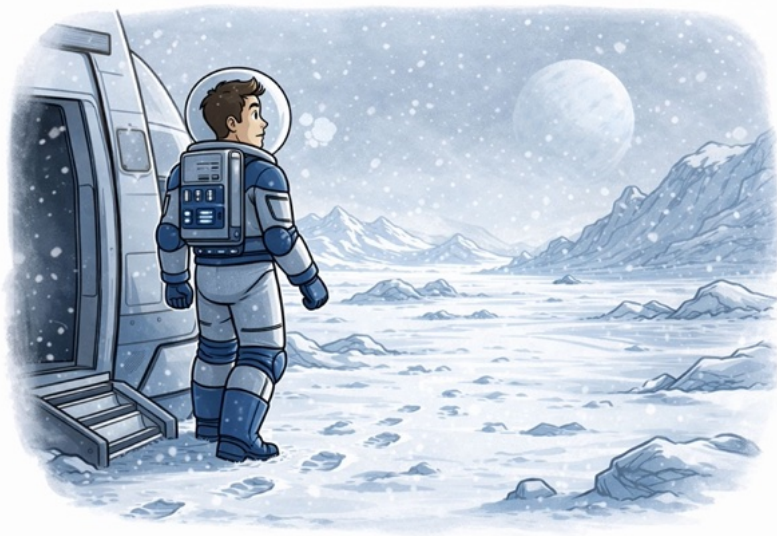
Objective prompt

Critique the attached sci-fi short story. Assign a score per category, then sum the scores at the end.

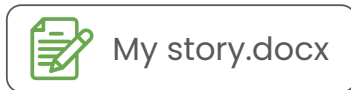


1. **Characters:** 10 / 25. Yoel wants to revive his family's legacy, but Cyrena's goal doesn't conflict.
2. **Plot:** 15 / 25 ...

Overall: 65 / 100. Great work, and keep fleshing out your characters!



Poorly written rubrics encourage sycophancy



Critique the attached sci-fi short story I wrote. Please **score it** out of 100.

Then assign a score per category: characters, plot, world-building, and writing craft (25 points each).



Scoring early
disrupts reasoning

Vague categories
invite subjectivity

💡 This type of rubric will
consistently give a higher score

Cross-model review



ChatGPT



Game files.pdf

Write a user manual for my fantasy RPG



Here's the user manual for *Legends of Aranthia* in a PDF.



User Manual.pdf



Gemini



Rubric.docx



User Manual.pdf

Critique the following game manual using this rubric.



My assessment:

1. Audience clarity: 15 points
2. Specificity: 25 points
3. Structural clarity: 20 points
4. Language and tone: 10 points

Total = 70 points

Trying different models keeps you sharp

**Different models have
different strengths**



**AI models have
jagged intelligence**

